

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]

Tap amplitudes (pixels): [122.41323457861898, 118.81077392223317, 110.43550153822818, 114.85643212288984, 107.07474025184464, 107.07474025184464, 104.04326023342406, 116.21101496846157, 109.85900054160332, 136.12494260788506, 98.71676655968832]



Based on the provided extracted features for MDS-UPDRS Item 3.4 (Finger Tapping), here is the analysis:

Analysis:

1. Rhythm and Interruptions (Tap Periods):

- The tap periods are: [150, 149, 154, 172, 156, 183, 138, 170, 197, 131] .
- The sequence shows significant variability (dysrhythmia). Comparing the periods to the approximate baseline/start (~150 frames):
 - Taps 1-3 are stable (~150).
 - Tap 4 slows to **172** (Interruption 1).
 - Tap 6 slows to **183** (Interruption 2).
 - Tap 8 slows to **170** (Interruption 3).
 - Tap 9 slows significantly to **197** (Interruption 4).
- There are approximately **4 distinct interruptions** or hesitations where the speed drops noticeably before recovering or fluctuating. This count aligns with the criteria for **Score 2 (Mild)**, which specifies "3 to 5 interruptions during tapping".

2. Amplitude (Decrement):

- The amplitudes are: [122.4, 118.8, 110.4, 114.9, 107.1, 107.1, 104.0, 116.2, 109.9, 136.1, 98.7] .
- There is a decreasing trend in the first half of the sequence (122.4 → 104.0 by the 7th point).
- The amplitude decrements midway through the sequence (around taps 5-7), further supporting a score of 2. While there are recoveries (e.g., to 116.2 and 136.1) towards the end, the initial sustained drop characterizes a decrement.

Conclusion: The subject demonstrates **Mild** impairment. The primary driver for this score is the presence of 3 to 5 interruptions in the tapping rhythm (specifically 4 hesitations) and a decrement in amplitude appearing midway through the sequence.